



This chapter gives a gentle yet concise introduction to most of the terminology used later in the book. Fortunately, much of standard graph theoretic terminology is so intuitive that it is easy to remember; the few terms better understood in their proper setting will be introduced later, when their time has come.

Section 1.1 offers a brief but self-contained summary of the most basic definitions in graph theory, those centred round the notion of a graph. Most readers will have met these definitions before, or will have them explained to them as they begin to read this book. For this reason, Section 1.1 does not dwell on these definitions more than clarity requires: its main purpose is to collect the most basic terms in one place, for easy reference later. For deviations for multigraphs see Section 1.10.

From Section 1.2 onwards, all new definitions will be brought to life almost immediately by a number of simple yet fundamental propositions. Often, these will relate the newly defined terms to one another: the question of how the value of one invariant influences that of another underlies much of graph theory, and it will be good to become familiar with this line of thinking early.

By $\mathbb{N}$ we denote the set of natural numbers, including zero. The set $\mathbb{Z}/n\mathbb{Z}$ of integers modulo n is denoted by $\mathbb{Z}_n$; its elements are written as $\bar{i} := i + n\mathbb{Z}$. When we regard $\mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$ as a field, we also denote it as $\mathbb{F}_2 = \{0, 1\}$. For a real number x we denote by $\lfloor x \rfloor$ the greatest integer $\leq x$, and by $\lceil x \rceil$ the least integer $\geq x$. Logarithms written as ‘log’ are taken at base 2; the natural logarithm will be denoted by ‘ln’. The expressions $x := y$ and $y =: x$ mean that x is being defined as y .

A set $\mathcal{A} = \{A_1, \dots, A_k\}$ of disjoint subsets of a set A is a *partition* of A if the union $\bigcup \mathcal{A}$ of all the sets $A_i \in \mathcal{A}$ is A . Our default assumption will be that all the A_i are non-empty. Another partition $\{A'_1, \dots, A'_\ell\}$ of A *refines* the partition $\mathcal{A}$ if each A'_i is contained in some A_j . By $[A]^k$ we denote the set of all k -element subsets of A . Sets with k elements will be called *k-sets*; subsets with k elements are *k-subsets*.

 $\mathbb{Z}_n$ $\lfloor x \rfloor, \lceil x \rceil$
log, ln

partition

 $[A]^k$ *k-set*

1.1 Graphs

graph A *graph* is a pair $G = (V, E)$ of sets such that $E \subseteq [V]^2$; thus, the elements of E are 2-element subsets of V . To avoid notational ambiguities, we shall always assume tacitly that $V \cap E = \emptyset$. The elements of V are the *vertices* (or *nodes*, or *points*) of the graph G , the elements of E are its *edges* (or *lines*). The usual way to picture a graph is by drawing a dot for each vertex and joining two of these dots by a line if the corresponding two vertices form an edge. Just how these dots and lines are drawn is considered irrelevant: all that matters is the information of which pairs of vertices form an edge and which do not.

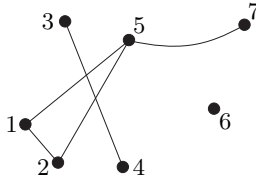


Fig. 1.1.1. The graph on $V = \{1, \dots, 7\}$ with edge set $E = \{\{1, 2\}, \{1, 5\}, \{2, 5\}, \{3, 4\}, \{5, 7\}\}$

on A graph with vertex set V is said to be a graph *on* V . The vertex set of a graph G is referred to as $V(G)$, its edge set as $E(G)$. These conventions are independent of any actual names of these two sets: the vertex set W of a graph $H = (W, F)$ is still referred to as $V(H)$, not as $W(H)$. We shall not always distinguish strictly between a graph and its vertex or edge set. For example, we may speak of a vertex $v \in G$ (rather than $v \in V(G)$), an edge $e \in G$, and so on.

order The number of vertices of a graph G is its *order*, written as $|G|$; its number of edges is denoted by $\|G\|$. Graphs are *finite*, *infinite*, *countable* and so on according to their order. Except in Chapter 8, our graphs will be finite unless otherwise stated.

$\emptyset$ For the *empty graph* $(\emptyset, \emptyset)$ we simply write $\emptyset$. A graph of order 0 or 1 is called *trivial*. Sometimes, e.g. to start an induction, trivial graphs can be useful; at other times they form silly counterexamples and become a nuisance. To avoid cluttering the text with non-triviality conditions, we shall mostly treat the trivial graphs, and particularly the empty graph $\emptyset$, with generous disregard.

incident ends A vertex v is *incident* with an edge e if $v \in e$; then e is an edge *at* v . The two vertices incident with an edge are its *endvertices* or *ends*, and an edge *joins* its ends. An edge $\{x, y\}$ is usually written as xy (or yx). If $x \in X$ and $y \in Y$, then xy is an X - Y edge. The set of all X - Y edges in a set E is denoted by $E(X, Y)$; instead of $E(\{x\}, Y)$ and $E(X, \{y\})$ we simply write $E(x, Y)$ and $E(X, y)$. The set of all the edges in E at a vertex v is denoted by $E(v)$.

Two vertices x, y of G are *adjacent*, or *neighbours*, if $\{x, y\}$ is an edge of G . Two edges $e \neq f$ are *adjacent* if they have an end in common. If all the vertices of G are pairwise adjacent, then G is *complete*. A complete graph on n vertices is a K^n ; a K^3 is called a *triangle*.

adjacent
neighbour
complete
 K^n

Pairwise non-adjacent vertices or edges are called *independent*. More formally, a set of vertices or of edges is *independent* if no two of its elements are adjacent. Independent sets of vertices are also called *stable*.

inde-
pendent

Let $G = (V, E)$ and $G' = (V', E')$ be two graphs. A map $\varphi: V \rightarrow V'$ is a *homomorphism* from G to G' if it preserves the adjacency of vertices, that is, if $\{\varphi(x), \varphi(y)\} \in E'$ whenever $\{x, y\} \in E$. Then, in particular, for every vertex x' in the image of φ its inverse image $\varphi^{-1}(x')$ is an independent set of vertices in G . If φ is bijective and its inverse φ^{-1} is also a homomorphism (so that $xy \in E \Leftrightarrow \varphi(x)\varphi(y) \in E'$ for all $x, y \in V$), we call φ an *isomorphism*, say that G and G' are *isomorphic*, and write $G \cong G'$. An isomorphism from G to itself is an *automorphism* of G .

homo-
morphism

isomorphic
 $\cong$

We do not normally distinguish between isomorphic graphs. Thus, we usually write $G = G'$ rather than $G \cong G'$, speak of *the* complete graph on 17 vertices, and so on. If we wish to emphasize that we are only interested in the isomorphism type of a given graph, we informally refer to it as an *abstract graph*.

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A class of graphs that is closed under isomorphism is called a *graph property*. For example, ‘containing a triangle’ is a graph property: if G contains three pairwise adjacent vertices then so does every graph isomorphic to G . A map taking graphs as arguments is called a *graph invariant* if it assigns equal values to isomorphic graphs. The number of vertices and the number of edges of a graph are two simple graph invariants; the greatest number of pairwise adjacent vertices is another.

property

invariant

We set $G \cup G' := (V \cup V', E \cup E')$ and $G \cap G' := (V \cap V', E \cap E')$. If $G \cap G' = \emptyset$, then G and G' are *disjoint*. If $V' \subseteq V$ and $E' \subseteq E$, then

$G \cup G'$

$G \cap G'$

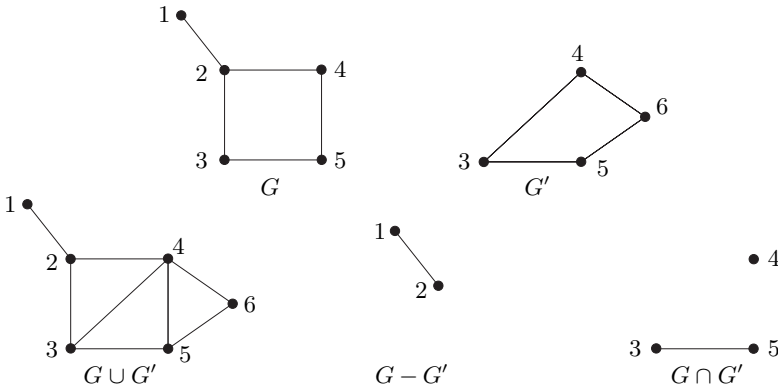


Fig. 1.1.2. Union, difference and intersection; the vertices 2,3,4 induce (or span) a triangle in $G \cup G'$ but not in G

subgraph
 $G' \subseteq G$

G' is a *subgraph* of G (and G a *supergraph* of G'), written as $G' \subseteq G$. Less formally, we say that G *contains* G' . If $G' \subseteq G$ and $G' \neq G$, then G' is a *proper subgraph* of G .

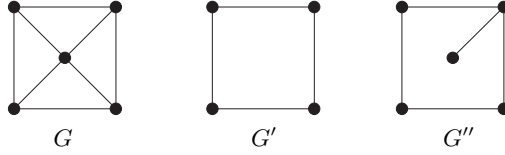


Fig. 1.1.3. A graph G with subgraphs G' and G'' :
 G' is an induced subgraph of G , but G'' is not

induced
subgraph

$G[U]$

spanning

–

+

edge-
maximal

minimal

maximal

$G * G'$

comple-
ment $\overline{G}$

line graph
 $L(G)$

If $G' \subseteq G$ and G' contains *all* the edges $xy \in E$ with $x, y \in V'$, then G' is an *induced subgraph* of G ; we say that V' *induces* or *spans* G' in G , and write $G' =: G[V']$. Thus if $U \subseteq V$ is any set of vertices, then $G[U]$ denotes the graph on U whose edges are precisely the edges of G with both ends in U . If H is a subgraph of G , not necessarily induced, we abbreviate $G[V(H)]$ to $G[H]$. Finally, $G' \subseteq G$ is a *spanning* subgraph of G if V' spans all of G , i.e. if $V' = V$.

If U is any set of vertices (usually of G), we write $G - U$ for $G[V \setminus U]$. In other words, $G - U$ is obtained from G by *deleting* all the vertices in $U \cap V$ and their incident edges. If $U = \{v\}$ is a singleton, we write $G - v$ rather than $G - \{v\}$. Instead of $G - V(G')$ we simply write $G - G'$. For a subset F of $[V]^2$ we write $G - F := (V, E \setminus F)$ and $G + F := (V, E \cup F)$; as above, $G - \{e\}$ and $G + \{e\}$ are abbreviated to $G - e$ and $G + e$. We call G *edge-maximal* with a given graph property if G itself has the property but no graph (V, F) with $F \supsetneq E$ does.

More generally, when we call a graph *minimal* or *maximal* with some property but have not specified any particular ordering, we are referring to the subgraph relation. When we speak of minimal or maximal sets of vertices or edges, the reference is simply to set inclusion.

If G and G' are disjoint, we denote by $G * G'$ the graph obtained from $G \cup G'$ by joining all the vertices of G to all the vertices of G' . For example, $K^2 * K^3 = K^5$. The *complement* $\overline{G}$ of G is the graph on V with edge set $[V]^2 \setminus E$. The *line graph* $L(G)$ of G is the graph on E in which $x, y \in E$ are adjacent as vertices if and only if they are adjacent as edges in G .

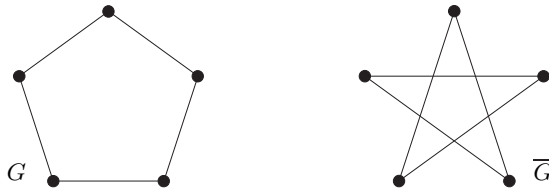


Fig. 1.1.4. A graph isomorphic to its complement

1.2 The degree of a vertex

Let $G = (V, E)$ be a (non-empty) graph. The set of neighbours of a vertex v in G is denoted by $N_G(v)$, or briefly by $N(v)$.¹ More generally for $U \subseteq V$, the neighbours in $V \setminus U$ of vertices in U are called *neighbours of U* ; their set is denoted by $N(U)$. $N(v)$

The *degree* (or *valency*) $d_G(v) = d(v)$ of a vertex v is the number *degree $d(v)$*
 $|E(v)|$ of edges at v ; by our definition of a graph,² this is equal to the number of neighbours of v . A vertex of degree 0 is *isolated*. *isolated*
The number $\delta(G) := \min \{ d(v) \mid v \in V \}$ is the *minimum degree* of G , the number $\delta(G)$
 $\Delta(G) := \max \{ d(v) \mid v \in V \}$ its *maximum degree*. If all the vertices $\Delta(G)$
of G have the same degree k , then G is *k -regular*, or simply *regular*. *regular*
A 3-regular graph is called *cubic*. *cubic*

The number

$$d(G) := \frac{1}{|V|} \sum_{v \in V} d(v) \quad d(G)$$

is the *average degree* of G . Clearly, *average degree*

$$\delta(G) \leq d(G) \leq \Delta(G).$$

The average degree quantifies globally what is measured locally by the vertex degrees: the number of edges of G per vertex. Sometimes it will be convenient to express this ratio directly, as $\varepsilon(G) := |E|/|V|$. $\varepsilon(G)$

The quantities d and ε are, of course, intimately related. Indeed, if we sum up all the vertex degrees in G , we count every edge exactly twice: once from each of its ends. Thus

$$|E| = \frac{1}{2} \sum_{v \in V} d(v) = \frac{1}{2} d(G) \cdot |V|,$$

and therefore

$$\varepsilon(G) = \frac{1}{2} d(G).$$

Proposition 1.2.1. *The number of vertices of odd degree in a graph is always even.* [10.3.1]

Proof. As $|E| = \frac{1}{2} \sum_{v \in V} d(v)$ is an integer, $\sum_{v \in V} d(v)$ is even. $\square$

¹ Here, as elsewhere, we drop the index referring to the underlying graph if the reference is clear.

² but not for multigraphs; see Section 1.10